

TB Scrambler

DETAILED ENGLISH USER MANUAL

A 31-band spectral permutation effect that rearranges a selected frequency region while leaving frequencies outside the range intact.



Catalog version: 2.3.0

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Host-visible endpoints documented: 5

1. Product purpose

A 31-band spectral permutation effect that rearranges a selected frequency region while leaving frequencies outside the range intact.

Full-band glitch and resynthesis can destroy rhythm, low-end weight, and mix placement. TB Scrambler confines its transformation between Low and High boundaries, then rearranges spectral structure inside that window so recognition can be reduced without discarding the whole source.

What result it creates

- Vocal and loop de-recognition
- Metallic, encrypted, and radio-like formants
- Protected bass and air outside the selected window
- Continuous movement from reconstruction to full displacement

Signal flow

Input -> 31-band analysis -> Low/High region selection -> repeatable spectral permutation -> Scramble Amount interpolation -> reconstruction

2. Complete feature guide

Low and High boundaries

Only spectral bands between Low and High are rearranged. Keep Low above the fundamental to preserve bass weight; lower High to retain natural air.

Scramble Amount

At 0 percent the selected region reconstructs normally. Increasing the value moves the formant envelope and fine structure toward the fixed permutation.

Repeatability

The permutation is repeatable rather than a new random map every block, which makes presets and automation stable.

Programs

Factory programs move from Vocal Haze and Drum Grit through Formant Flip, Alien Broadcast, Glass Shatter, and Total Dislocation.

3. Quick start

1. Set Scramble Amount to 100 percent temporarily.
2. Move Low and High until only the intended spectrum is transformed.
3. Reduce Scramble Amount to the usable intensity.
4. Compare with bypass at matched loudness.
5. Automate Amount for transitions and boundaries for evolving focus.

4. Practical workflows

Alien vocal

1. Set Low above the vocal fundamental.
2. Set High around the upper intelligibility region.
3. Raise Scramble Amount until formants become synthetic.
4. Back Amount down if words need to remain clear.

Expected result: The voice becomes encrypted and metallic while low-end presence survives.

Drum texture

1. Protect the kick below Low.
2. Limit High to keep cymbal air natural.
3. Use a medium Amount and automate toward full displacement for fills.

Expected result: The midrange becomes fractured without erasing the groove.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Low must remain below High for a useful processing window.
- High Amount can make speech unintelligible by design.
- Latency and spectral smearing are normal consequences of band analysis.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 5 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
High VST3 ID 3202466	20 to 20000	4500	Automatable	Host-automatable control for High. Its full range and default are shown in this table; use it with the related feature section above.
Low VST3 ID 107348	20 to 20000	300	Automatable	Host-automatable control for Low. Its full range and default are shown in this table; use it with the related feature section above.
Program VST3 ID 1886548852	Init, Vocal Haze, Drum Grit, Bass Wobble, Mid Fracture, Radio Scramble, Yolk Crumble, Formant Flip, Wide Disorder, Alien Broadcast, Glass Shatter, Total Dislocation	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Scramble Amount VST3 ID 733630552	0.0 to 100.0	50.0	Automatable	Sets how strongly the named process affects the signal.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.